HAT-P-36b

R-1495

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_210528` · created 2026-08-01T14:12:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459362.7242 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.031
Transit depth (Rp/Rs)^2	0.88 +/- 0.59 [%]
Orbital Inclination (inc)	89.22 +/- 2.8
Airmass coefficient 1 (a1)	1.01 +/- 0.016
Airmass coefficient 2 (a2)	-0.006 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#2 - [92, 54]
Optimal Aperture	3.68
Optimal Annulus	8.47
Transit Duration (day)	0.093 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.094 ± 0.031 vs 0.126 (deviation 0.92σ)
Duration fit vs published	0.093 d vs 0.0925 d (deviation 0.11σ)
Mid-time O–C	4.4 min
Depth SNR	2.7
Residual scatter	1.55 %

Light curve

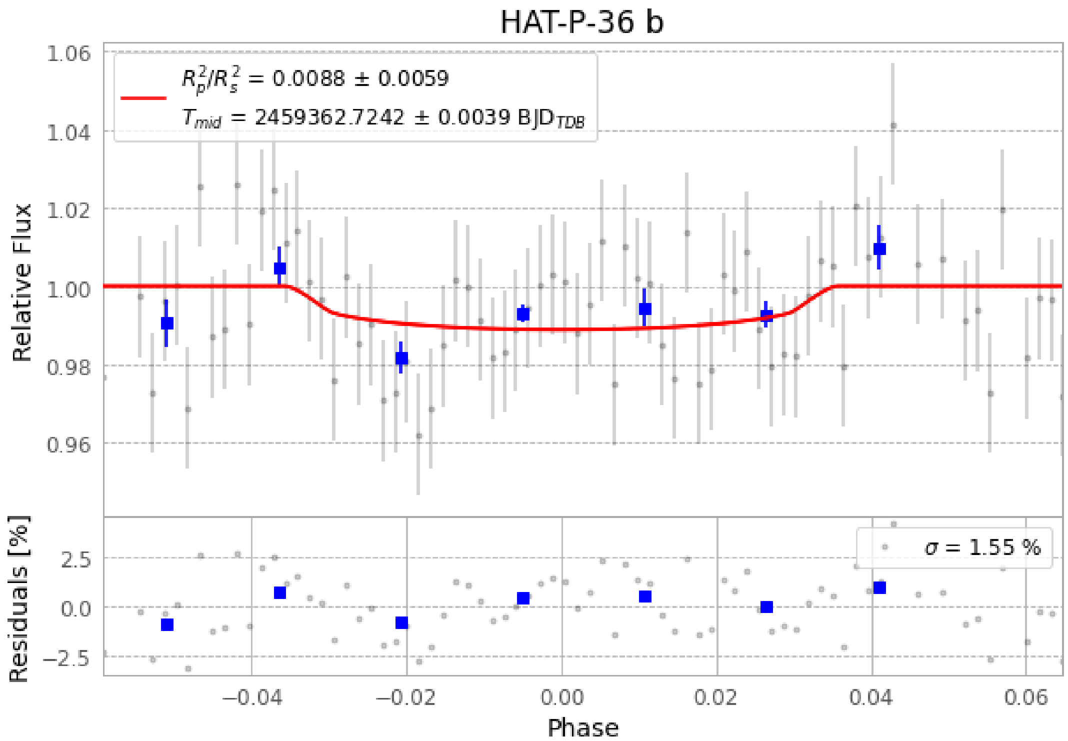

Figure 1 — FinalLightCurve_HAT-P-36 b_2021-05-27.png (SHA-256 ee431f3392670b6a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 219], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 21a70475f1ca57b6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

78 FITS frames (51 MB), `HATP-36210528032709.FITS` → `HATP-36210528072412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210528.log (8c116f1bd7a1…), FinalLightCurve_HAT-P-36 b_2021-05-27.png (ee431f339267…), FinalLightCurve_HAT-P-36 b_2021-05-27.csv (1860e33c232d…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 14:12Z.